

Improved and Restricted Bounds for Off-Diagonal Unordered Erdős–Rado Numbers

Graph Seminar Notes – Referee-audited working version

8 October 2026

Status of the statements in these notes. The proofs of Theorems 1.3, 5.2, and 6.2, and of the local sparsification Lemma 4.1, have been rewritten after a referee-style audit of the conflict counts, random sparsification, high-degree pruning, and the Duke–Lefmann–Rödl hypothesis. The literature discussion in Section 8 records the closest precedents found in a broad public-literature search as of 8 October 2026. Absence of a located precedent is not an absolute bibliographic guarantee.

Abstract

For fixed integers m, ℓ and $r \rightarrow \infty$, let $\text{ER}(m, \ell, r)$ be the least n such that every edge-colouring of K_n contains a monochromatic K_m , a lexical K_ℓ , or a rainbow K_r . Araujo and Peng proved $\text{ER}(4, 4, r) = O(r^7/(\log r)^2)$. We record a logarithmic improvement

$$\text{ER}(4, 4, r) = O\left(\frac{r^7}{(\log r)^3}\right),$$

obtained by retaining the distinguished colour inside a large monochromatic neighbourhood and exploiting the resulting triangle-free graph with Shearer’s independence bound.

We then isolate a parameterized version of the Alon–Jiang–Miller–Pritikin/Duke–Lefmann–Rödl sparsification mechanism with one exceptional colour. Its first application gives a book-number interpolation theorem: if the maximum number β of monochromatic triangles sharing a monochromatic edge satisfies $\beta \leq r^{2-\varepsilon}$, then the conjectured scale $O_{m,\varepsilon}(r^6/(\log r)^2)$ holds for $\text{ER}(m, 4, r)$ inside this restricted class. A second structural reduction shows that the Araujo–Peng conjectured exponent for general ℓ holds for colourings whose monochromatic self-neighbourhood graphs have uniformly bounded fractional chromatic number.

A very recent preprint of Gvozdić, Hunter, Milojević and Sudakov disproves the unrestricted Araujo–Peng Conjecture 19 (and consequently the $\ell = 4$ conjecture for sufficiently large fixed m), while leaving the smallest case $\text{ER}(4, 4, r)$ unresolved. Thus the restricted results below should be read as identifying structural regimes in which the previously conjectured polynomial scale survives.

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1 Setup and theorem map

All logarithms are natural. Constants implicit in $O(\cdot), \Omega(\cdot)$ may depend on fixed parameters such as m, ℓ, q, ε .

Definition 1.1 (Lexical colouring). An edge-colouring of a clique on ordered vertices $v_1, \dots, v_s$ is *lexical* if two edges have the same colour if and only if they have the same lower endpoint. Thus the $s - 1$ colour classes are the stars rooted at $v_1, \dots, v_{s-1}$ and their colours are pairwise distinct.

Definition 1.2 (Off-diagonal unordered Erdős–Rado number). $\text{ER}(m, \ell, r)$ is the minimum n such that every edge-colouring of K_n contains a monochromatic K_m , a lexical K_ℓ , or a rainbow K_r .

Write

$$A_r := \frac{r^3}{\log r}.$$

Araujo–Peng proved

$$\text{ER}(m, 3, r) = O_m(A_r), \quad \text{ER}(3, \ell, r) = \Theta_\ell(A_r^{\ell-2}),$$

and

$$\text{ER}(4, 4, r) = O\left(\frac{r^7}{(\log r)^2}\right).$$

The following are the three results developed in these notes.

Theorem 1.3 (Unconditional logarithmic improvement). *There is an absolute constant $C > 0$ such that, for all sufficiently large r ,*

$$\boxed{\text{ER}(4, 4, r) \leq C \frac{r^7}{(\log r)^3}}.$$

Theorem 1.4 (Book-number interpolation; informal form). *Fix $m \geq 4$. Let χ be an edge-colouring and let*

$$\beta(\chi) := \max_{xy} |N_{\chi(xy)}(x) \cap N_{\chi(xy)}(y)|.$$

If $\beta(\chi) \leq \beta \leq r^2$, then a complete graph of order

$$O_m \left(\frac{r^6}{\log r [1 + \log(r^2 / \max\{1, \beta\})]} \right)$$

forces a monochromatic K_m , a lexical K_4 , or a rainbow K_r . In particular, if $\beta \leq r^{2-\varepsilon}$, the bound becomes

$$O_{m,\varepsilon} \left(\frac{r^6}{(\log r)^2} \right).$$

Theorem 1.5 (Fractional self-neighbourhood reduction; informal form). *Fix $m, \ell \geq 3$ and $\kappa \geq 1$. In the class of colourings for which every monochromatic self-neighbourhood graph has fractional chromatic number at most κ ,*

$$\text{ER}_\kappa(m, \ell, r) \leq C_{m,\ell} \kappa^{\ell-3} A_r^{\ell-2}.$$

Updated conjectural context. Araujo–Peng Conjecture 19 asserted $\text{ER}(m, \ell, r) = O_{m,\ell}(A_r^{\ell-2})$ for every fixed $m, \ell \geq 4$. Gvozdić–Hunter–Milojević–Sudakov (2026) proved lower bounds $\text{ER}(a, b, c) \geq c^{\Omega_\eta(ab)}$ in the regime $c \geq a^{1+\eta}$, and explicitly noted that this disproves Conjecture 19. Taking $b = 4$ also rules out the corresponding $O_m(c^6/(\log c)^2)$ assertion for all sufficiently large fixed m . The case $m = 4$, namely

$$\text{ER}(4, 4, r) \stackrel{?}{=} O \left(\frac{r^6}{(\log r)^2} \right),$$

remains the principal small case addressed by Theorem 1.3 and the book-restricted theorem.

2 Inputs from the literature

We use three classical ingredients.

Theorem 2.1 (AJMP rainbow-clique bound [1]). *There is an absolute constant C such that every Δ -good edge-colouring of K_n contains a rainbow K_r whenever*

$$n \geq C \Delta \frac{r^3}{\log r}.$$

Here Δ -good means that every colour class has maximum degree at most Δ .

Theorem 2.2 (Shearer [8]). *If G is triangle-free on s vertices with average degree d , then*

$$\alpha(G) \geq s \frac{d \log d - d + 1}{(d - 1)^2}.$$

In particular, for an absolute $c > 0$,

$$\alpha(G) \geq cs \frac{\log(d + 2)}{d + 1}.$$

Theorem 2.3 (Duke–Lefmann–Rödl, in the form used by AJMP [3, 1]). *For every fixed $k \geq 3$ there is $a_k > 0$ such that the following holds. Let $\mathcal{H}$ be a k -uniform hypergraph on s vertices, with no 2-cycles (equivalently, any two distinct edges meet in at most one vertex), and*

$$\Delta(\mathcal{H}) \leq b^{k-1}, \quad b \geq b_0(k).$$

Then

$$\alpha(\mathcal{H}) \geq a_k \frac{s}{b} (\log b)^{1/(k-1)}.$$

We also repeatedly use the following two localisation statements from Araujo–Peng [2].

Lemma 2.4 (Araujo–Peng local bounds). *Suppose an edge-colouring contains neither a monochromatic K_m nor a lexical K_4 . Fix v and a colour a , and put $V \subseteq N_a(v)$. Then for each $u \in V$ and each colour $c \neq a$,*

$$d_c(u; V) < R(m-1).$$

If, in addition, $m = 4$ and the colouring contains no rainbow K_r , then

$$d_a(u; V) < \text{ER}(4, 3, r) = O(A_r).$$

For $m = 4$, the graph $G_a[V]$ is triangle-free.

Proof. The first assertion is Claim 16 of [2]: if $u \in N_a(v)$ and $d_c(u; V) \geq R(m-1)$ for $c \neq a$, then either a third colour occurs inside $N_c(u) \cap N_a(v)$, producing a lexical K_4 with u, v , or the induced two-colouring yields a monochromatic K_{m-1} , which extends with u or v to a monochromatic K_m . The second assertion is Claim 17 with $m = \ell = 4$. Finally, a triangle of colour a inside $N_a(v)$ together with v is a monochromatic K_4 . $\square$

3 The unconditional improvement for $\text{ER}(4, 4, r)$

The new point is that the distinguished colour should not be deleted. It induces a triangle-free graph, and Shearer’s theorem recovers an additional factor of $\log r$.

3.1 Conflict counts

Fix a set $V \subseteq N_a(v)$ of size N in a colouring with no monochromatic K_4 , lexical K_4 , or rainbow K_r . Relabel $a = 1$. By Lemma 2.4,

$$d_c(u; V) \leq 5 \quad (c \neq 1), \quad d_1(u; V) \leq D_0 := C_0 A_r.$$

Let $X(S)$ count non-1 monochromatic copies of $K_{1,2}$ in S , and let $Y(S)$ count non-1 monochromatic copies of $2K_2$. The counting in [2] gives

$$X(V) \leq 2N^2, \quad Y(V) \leq \frac{5}{8}N^3. \tag{1}$$

Proof of Theorem 1.3. Choose

$$N := K \frac{r^4}{(\log r)^2}, \quad n := C_1 A_r N,$$

where K and then C_1 are sufficiently large absolute constants. Consider an arbitrary colouring of K_n avoiding the three target configurations.

If the colouring is N -good, Theorem 2.1 yields a rainbow K_r , a contradiction. Hence some vertex v has more than N neighbours in one colour, say colour 1. Fix an N -set $V \subseteq N_1(v)$.

Let

$$t := \lfloor \eta N^{1/3} \rfloor$$

for a sufficiently small fixed $\eta > 0$, and choose a uniformly random t -set $T \subseteq V$. By (1),

$$\mathbb{E}X(T) \leq 2N^2 \left(\frac{t}{N} \right)^3 = \frac{2t^3}{N} = O(\eta^3),$$

and

$$\mathbb{E}Y(T) \leq \frac{5}{8}N^3 \left(\frac{t}{N}\right)^4 = \frac{5t^4}{8N} = O(\eta^3 t).$$

Thus, after taking η small and r large,

$$\mathbb{E}[X(T) + Y(T)] \leq t/100. \quad (2)$$

Let $G := G_1[V]$. It is triangle-free and $\Delta(G) \leq D_0$, so $e(G) \leq D_0 N/2$. Therefore

$$\mathbb{E}e(G[T]) \leq \frac{D_0 t^2}{2N} (1 + o(1)). \quad (3)$$

By Markov's inequality and a union bound, there is a choice of T satisfying simultaneously

$$X(T) + Y(T) \leq t/4, \quad e(G[T]) \leq C_2 \frac{D_0 t^2}{N}.$$

Delete one vertex from every non-1 conflict counted by $X(T) + Y(T)$. We obtain $W \subseteq T$ with

$$|W| \geq 3t/4,$$

such that every non-1 colour occurs on at most one edge of K_W . The graph $G[W]$ is still triangle-free, and its average degree is at most

$$d_W \leq C_3 \frac{D_0 t}{N} =: d_*.$$

Since $x \mapsto \log(x+2)/(x+1)$ is decreasing for $x \geq 0$, Theorem 2.2 gives

$$\alpha(G[W]) \geq c|W| \frac{\log(d_* + 2)}{d_* + 1}.$$

For our parameters,

$$d_* \asymp D_0 N^{-2/3} = O\left(K^{-2/3} r^{1/3} (\log r)^{1/3}\right),$$

while

$$\frac{N}{D_0} \asymp K \frac{r}{\log r}.$$

Hence, for all sufficiently large r ,

$$\alpha(G[W]) \geq c' \frac{N}{D_0} \log r \geq c'' K r.$$

Choosing K large gives an independent r -set $R \subseteq W$ in the special-colour graph. There are no colour-1 edges in R , and every other colour occurs at most once; hence K_R is rainbow, a contradiction. Therefore $n = O(r^7/(\log r)^3)$. $\square$

Remark 3.1 (Where the extra logarithm comes from). The previous $O(r^7/(\log r)^2)$ argument deletes special-colour repetitions. Keeping the special colour instead produces a triangle-free graph and replaces a crude deletion cost by the factor $\log d/d$ from Shearer. At the relevant scale this contributes exactly one additional factor of $\log r$.

4 A one-exceptional-colour sparsification lemma

We now extract and parameterize the AJMP conflict-hypergraph mechanism. The technology is classical; the feature relevant here is that one colour is allowed to be arbitrary (up to an average-degree constraint), while all other colours are locally bounded.

Lemma 4.1 (Local sparsification with one exceptional colour). *For every fixed integer $q \geq 1$ there exist constants $c_q, C_q > 0$ and $B_0(q)$ such that the following holds for all sufficiently large N . Let χ be an edge-colouring of K_N with distinguished colour a . Assume*

$$\Delta(G_c) \leq q \quad (c \neq a),$$

and let the special-colour graph G_a have average degree d . Suppose there is a real B such that

$$B_0(q) \leq B \leq c_q N^{1/15}, \quad dB \leq c_q N^{2/3}. \quad (4)$$

Then χ contains a rainbow clique of order at least

$$\boxed{c_q N^{1/3} (\log B)^{1/3}}.$$

4.1 The conflict hypergraphs

Let $\mathcal{H}_3$ be the 3-uniform hypergraph of vertex triples spanning a non- a monochromatic $K_{1,2}$, and let $\mathcal{H}_4$ be the 4-uniform hypergraph of 4-sets containing a non- a monochromatic $2K_2$.

Proposition 4.2 (Basic conflict counts). *Under the hypotheses of Lemma 4.1,*

$$|\mathcal{H}_3| = O_q(N^2), \quad |\mathcal{H}_4| = O_q(N^3).$$

If μ_j is the number of unordered pairs of distinct edges of $\mathcal{H}_4$ meeting in exactly j vertices, then

$$\mu_2 = O_q(N^4), \quad \mu_3 = O_q(N^3).$$

Proof. For $\mathcal{H}_3$,

$$|\mathcal{H}_3| \leq \sum_{x \in V} \sum_{c \neq a} \binom{d_c(x)}{2} \leq \frac{q-1}{2} \sum_x \sum_{c \neq a} d_c(x) = O_q(N^2).$$

Writing $e_c = |E(G_c)|$, we have $e_c \leq qN/2$ and $\sum_c e_c \leq \binom{N}{2}$, whence

$$|\mathcal{H}_4| \leq \sum_{c \neq a} \binom{e_c}{2} = O_q(N^3).$$

Fix a pair $S = \{x, y\}$. We claim that the number $d_2(S)$ of edges of $\mathcal{H}_4$ containing S is $O_q(N)$. If xy itself is one edge of a witnessing monochromatic matching, then the other edge has colour $\chi(xy)$ and there are at most $qN/2$ choices. Otherwise x and y lie on different matching edges. For a prescribed pairing, the number of choices is at most

$$\sum_{c \neq a} d_c(x) d_c(y) \leq q \sum_{c \neq a} d_c(x) \leq qN,$$

and there are only two pairings. Thus $d_2(S) = O_q(N)$. Consequently

$$\mu_2 \leq \sum_{S \in \binom{V}{2}} \binom{d_2(S)}{2} = O_q(N^4).$$

Similarly, fix a triple $S = \{x, y, z\}$. A bad $2K_2$ on $S \cup \{w\}$ has one of three pairings. Once the edge inside S is chosen, the colour is fixed and the remaining vertex of S has at most q possible partners w in that colour. Hence $d_3(S) \leq 3q$ and

$$\mu_3 \leq \sum_{S \in \binom{V}{3}} \binom{d_3(S)}{2} = O_q(N^3).$$

□

4.2 Random sparsification, deletion, and pruning

Proof of Lemma 4.1. Set

$$p := \sigma_q \frac{B}{N^{2/3}},$$

where $\sigma_q > 0$ is a sufficiently small constant. Select each vertex independently with probability p , obtaining a random set Y .

The expected number of selected vertices is pN . For the special colour,

$$\mathbb{E}e(G_a[Y]) = p^2 e(G_a) \leq \frac{1}{2} dN p^2,$$

so

$$\frac{\mathbb{E}e(G_a[Y])}{pN} \leq \frac{dp}{2} = O_q\left(\frac{dB}{N^{2/3}}\right). \quad (5)$$

By (4), this is a sufficiently small constant.

For the 3-conflicts,

$$\frac{\mathbb{E}|\mathcal{H}_3[Y]|}{pN} = O_q(Np^2) = O_q(B^2 N^{-1/3}) = o(1).$$

For a pair of 4-edges meeting in two vertices, the union has six vertices, and hence

$$\frac{\mathbb{E}\mu_2(Y)}{pN} = O_q(N^3 p^5) = O_q(B^5 N^{-1/3}).$$

The restriction $B \leq c_q N^{1/15}$ makes this as small as desired. For intersection size three, the union has five vertices, so

$$\frac{\mathbb{E}\mu_3(Y)}{pN} = O_q(N^2 p^4) = O_q(B^4 N^{-2/3}) = o(1).$$

Finally,

$$\mathbb{E}|\mathcal{H}_4[Y]| = O_q(N^3 p^4).$$

A Chernoff bound gives $|Y| \geq pN/2$ with probability $1 - o(1)$. Apply Markov's inequality to the five nonnegative random variables

$$e(G_a[Y]), \quad |\mathcal{H}_3[Y]|, \quad \mu_2(Y), \quad \mu_3(Y), \quad |\mathcal{H}_4[Y]|,$$

with fixed sufficiently generous multiples of their expectations. A union bound gives a realization Y for which all these bounds hold simultaneously and $|Y| \geq pN/2$.

Delete one vertex from every special-colour edge, every edge of $\mathcal{H}_3[Y]$, and every 2-cycle of $\mathcal{H}_4[Y]$. Because the first four error quantities above are a sufficiently small fraction of pN , the resulting set Y_1 satisfies

$$|Y_1| \geq c_q pN.$$

Moreover,

$$G_a[Y_1] \text{ is empty,} \quad \mathcal{H}_3[Y_1] \text{ is empty,}$$

and $\mathcal{H}_4[Y_1]$ has no 2-cycles. Also

$$|\mathcal{H}_4[Y_1]| \leq C_q N^3 p^4.$$

Therefore its average hypergraph degree is at most

$$\frac{4|\mathcal{H}_4[Y_1]|}{|Y_1|} \leq C'_q N^2 p^3.$$

Delete every vertex whose degree in $\mathcal{H}_4[Y_1]$ exceeds twice this threshold. At most half the vertices are deleted, so an induced subhypergraph on a set Z remains with

$$|Z| \geq c'_q pN, \quad \Delta(\mathcal{H}_4[Z]) \leq C''_q N^2 p^3.$$

Put

$$b^3 := C''_q N^2 p^3.$$

Then $b = \Theta_q(B)$. Enlarging $B_0(q)$ ensures $b \geq b_0(4)$ and also $\log b \asymp_q \log B$. Theorem 2.3 yields

$$\alpha(\mathcal{H}_4[Z]) \geq c_q \frac{|Z|}{b} (\log b)^{1/3} \geq c_q \frac{pN}{B} (\log B)^{1/3} = c_q N^{1/3} (\log B)^{1/3}.$$

Let R be such an independent set. Colour a does not occur on K_R . If a non- a colour were repeated on two intersecting edges, the three involved vertices would form an edge of $\mathcal{H}_3$; if it were repeated on disjoint edges, their four vertices would form an edge of $\mathcal{H}_4$. Both are impossible. Thus R is rainbow. $\square$

Remark 4.3 (Origin of the exponent $1/15$). The bottleneck is the μ_2 term. A 2-cycle formed by two 4-edges with two common vertices occupies six vertices, hence survives with probability p^6 . Since $\mu_2 = O_q(N^4)$, normalizing by the target size pN produces

$$N^3 p^5 \asymp B^5 N^{-1/3}.$$

Thus the present argument requires $B^5 \ll N^{1/3}$, i.e. $B \ll N^{1/15}$.

5 Book-number interpolation

Definition 5.1 (Monochromatic book number). For an edge-colouring χ , define

$$\beta(\chi) := \max_{xy \in E(K_n)} |N_{\chi(xy)}(x) \cap N_{\chi(xy)}(y)|.$$

Thus $\beta(\chi)$ is the maximum number of monochromatic triangles sharing a monochromatic edge.

For clarity, let $\text{ER}_{\leq \beta}^{\text{book}}(m, 4, r)$ denote the least n such that every colouring of K_n with $\beta(\chi) \leq \beta$ contains a monochromatic K_m , a lexical K_4 , or a rainbow K_r .

Theorem 5.2 (Book-number interpolation). *Fix $m \geq 4$. For $1 \leq \beta \leq r^2$, put*

$$L_\beta := 1 + \log \frac{r^2}{\beta}.$$

Then

$$\text{ER}_{\leq \beta}^{\text{book}}(m, 4, r) = O_m \left(\frac{r^6}{\log r L_\beta} \right).$$

The same conclusion holds with β replaced everywhere by $\max\{1, \beta\}$ if one allows $\beta < 1$ as a real parameter.

Proof. Let

$$N := K_m \frac{r^3}{L_\beta}, \quad n := C_m A_r N,$$

with K_m, C_m sufficiently large. Consider a colouring of K_n satisfying the book restriction and avoiding a monochromatic K_m , lexical K_4 , and rainbow K_r .

If the colouring is N -good, Theorem 2.1 gives a rainbow K_r . Hence some vertex v has at least N neighbours in a colour a . Fix an N -set $V \subseteq N_a(v)$.

Lemma 2.4 gives, for every $c \neq a$,

$$\Delta(G_c[V]) \leq q_m := R(m-1) - 1.$$

For the distinguished colour, every $u \in V$ satisfies

$$d_{G_a[V]}(u) \leq |N_a(u) \cap N_a(v)| \leq \beta,$$

so the average degree d of $G_a[V]$ is at most β .

Choose

$$B := c_m \min \left\{ N^{1/15}, \frac{N^{2/3}}{\beta} \right\}$$

(with the harmless replacement $\beta \mapsto \max\{1, \beta\}$ when necessary). Then $dB \leq c_m N^{2/3}$ and $B \leq c_m N^{1/15}$. It remains to verify the logarithmic size of B . Since $L_\beta \leq 1 + 2 \log r$,

$$\log N^{1/15} = \frac{1}{15} (3 \log r - \log L_\beta + O_m(1)) \geq c L_\beta$$

for an absolute $c > 0$ after decreasing c and taking r large. Also

$$\log \frac{N^{2/3}}{\beta} = \log \frac{r^2}{\beta} - \frac{2}{3} \log L_\beta + O_m(1) \geq c L_\beta,$$

with the bounded- L_β range absorbed by choosing K_m sufficiently large. Thus

$$\log B \geq c'_m L_\beta,$$

and $B \geq B_0(q_m)$.

Lemma 4.1 now gives a rainbow clique of order at least

$$c_m N^{1/3} (\log B)^{1/3} \geq c_m \left(K_m \frac{r^3}{L_\beta} \right)^{1/3} L_\beta^{1/3} = c_m K_m^{1/3} r.$$

Taking K_m large makes this at least r , a contradiction. $\square$

Corollary 5.3 (Power-saving book restriction). *For fixed $m \geq 4$ and $\varepsilon > 0$, if $\beta(\chi) \leq r^{2-\varepsilon}$, then*

$$\text{ER}_{\leq r^{2-\varepsilon}}^{\text{book}}(m, 4, r) = O_{m,\varepsilon} \left(\frac{r^6}{(\log r)^2} \right).$$

Proof. Now $L_\beta \geq \varepsilon \log r + O(1)$, so Theorem 5.2 gives the result. $\square$

Corollary 5.4 (Polylogarithmic saving). *If $\beta \leq r^2/(\log r)^K$ for fixed $K > 0$, then*

$$\text{ER}_{\leq \beta}^{\text{book}}(m, 4, r) = O_{m,K} \left(\frac{r^6}{\log r \log \log r} \right).$$

Remark 5.5 (Interpretation after the 2026 counterexample). The unrestricted Araujo–Peng $\ell = 4$ conjecture is false for all sufficiently large fixed m , by the new lower-bound regime of Gvozdić–Hunter–Milojević–Sudakov. Corollary 5.3 therefore identifies a broad structural class in which the formerly conjectured $r^6/(\log r)^2$ scale is restored. For $m = 4$ it remains a genuine partial result toward the still-open smallest case.

6 A fractional-chromatic self-neighbourhood reduction

Definition 6.1 (Self-neighbourhood complexity). For a colouring χ and a vertex-colour pair (v, a) , let

$$H_{v,a} := G_a[N_a(v)].$$

For $\kappa \geq 1$, let $\mathcal{C}_\kappa$ be the class of colourings satisfying

$$\chi_f(H_{v,a}) \leq \kappa \quad \text{for every } (v, a).$$

Let $\text{ER}_\kappa(m, \ell, r)$ be the corresponding restricted Erdős–Rado number.

Theorem 6.2 (Bounded fractional self-neighbourhoods). *For fixed integers $m, \ell \geq 3$ and $\kappa \geq 1$,*

$$\text{ER}_\kappa(m, \ell, r) \leq C_{m,\ell,\kappa} \ell^{-3} \left(\frac{r^3}{\log r} \right)^{\ell-2}.$$

Proof. Write $F_\ell := \text{ER}_\kappa(m, \ell, r)$. The base case follows from Araujo–Peng:

$$F_3 \leq \text{ER}(m, 3, r) = O_m(A_r).$$

Assume $\ell \geq 4$ and the assertion holds for $\ell - 1$. Put

$$\Delta := \lceil \kappa F_{\ell-1} \rceil.$$

Take

$$n \geq C \Delta A_r.$$

If the colouring is Δ -good, Theorem 2.1 gives a rainbow K_r . Otherwise there are v, a with

$$|N_a(v)| > \kappa F_{\ell-1}.$$

Since $\chi_f(H_{v,a}) \leq \kappa$ and $\chi_f(H) \geq |V(H)|/\alpha(H)$ for every graph H ,

$$\alpha(H_{v,a}) \geq \frac{|N_a(v)|}{\kappa} > F_{\ell-1}.$$

Choose an independent set $W \subseteq N_a(v)$ of size greater than $F_{\ell-1}$. Thus no edge inside W has colour a .

The class $\mathcal{C}_\kappa$ is hereditary under vertex deletion: for an induced subcolouring on W , every new self-neighbourhood graph is an induced subgraph of its old counterpart, and fractional chromatic number is monotone under induced subgraphs. Hence the induction hypothesis applies inside W .

If it yields a monochromatic K_m or a rainbow K_r , we are done. Otherwise W contains a lexical $K_{\ell-1}$, say on ordered vertices $w_1, \dots, w_{\ell-1}$. None of its internal colours is a . Prepending v gives the ordering

$$v, w_1, \dots, w_{\ell-1},$$

and every edge from v to a w_i has colour a , while the internal lexical star-colours remain pairwise distinct and different from a . Thus these vertices form a lexical K_ℓ .

Consequently

$$F_\ell \leq C\kappa A_r F_{\ell-1},$$

and iteration from the base case gives

$$F_\ell \leq C_{m,\ell} \kappa^{\ell-3} A_r^{\ell-2}.$$

□

Corollary 6.3 (Bipartite self-neighbourhoods). *If every $H_{v,a}$ is bipartite, then $\kappa \leq 2$, and therefore*

$$\text{ER}_{\text{bip}}(m, \ell, r) = O_{m,\ell}(A_r^{\ell-2}).$$

Remark 6.4. Unlike the unrestricted Araujo–Peng Conjecture 19, which is now known to be false, Theorem 6.2 shows that its proposed exponent is valid on every bounded local-fractional-complexity class. The proof is elementary once the correct local parameter is identified; its role is structural rather than a new theorem in fractional colouring theory.

7 What remains for $\text{ER}(4, 4, r)$

The results isolate two obstructions to the conjectured $O(r^6/(\log r)^2)$ scale in the smallest case.

Inside a large monochromatic neighbourhood $V \subseteq N_a(v)$ for $m = 4$:

- (i) $G_a[V]$ is triangle-free;
- (ii) every non- a colour has maximum degree at most 5;
- (iii) the book theorem handles the regime in which the special-colour book number has a fixed power saving below r^2 ;
- (iv) the fractional reduction handles regimes in which the self-neighbourhood graph has uniformly bounded fractional chromatic number.

Thus a counterexample at the conjectured scale must simultaneously exhibit high book degree and substantial fractional-chromatic complexity inside triangle-free monochromatic self-neighbourhoods, while all non-special colours remain locally bounded.

For Lemma 4.1, the most visible quantitative bottleneck is

$$\mu_2 = O_q(N^4).$$

Any ER-specific improvement of the form $\mu_2 = O(N^{4-\delta})$ on the relevant localized configuration would enlarge the admissible range of B and is a concrete direction for further work.

8 Novelty audit and closest precedents

This section records a broad search of publicly indexed literature completed on 8 October 2026. Searches included exact expressions for $\text{ER}(4, 4, r)$ and its logarithmic exponents; “one exceptional colour” plus locally bounded/rainbow clique terminology; monochromatic book/book-number plus canonical Ramsey terminology; fractional chromatic/self-neighbourhood plus canonical Ramsey terminology; recent papers citing or discussing Araujo–Peng; and the principal locally bounded rainbow-colouring literature.

8.1 Theorem A

Finding: high-confidence novelty. No source located in the search gives

$$\text{ER}(4, 4, r) = O(r^7/(\log r)^3)$$

or a stronger bound for this fixed case. More significantly, the 2 October 2026 preprint of Gvozdić–Hunter–Milojević–Sudakov still lists the Araujo–Peng bound

$$\text{ER}(4, 4, c) = O(c^7/\log^2 c)$$

as the known result in this regime [4]. This is strong contemporaneous evidence that Theorem 1.3 is not already in the standard off-diagonal canonical-Ramsey literature.

8.2 The one-exceptional-colour Lemma 4.1

Finding: no statement-level precedent located; method-level precedent is AJMP. AJMP [1] already introduces the 3-uniform bad-2-claw and 4-uniform bad- $2K_2$ conflict systems, proves the same orders

$$|\mathcal{H}_3| = O(mN^2), \quad |\mathcal{H}_4| = O(mN^3), \quad \mu_2 = O(m^2N^4), \quad \mu_3 = O(m^2N^3),$$

randomly sparsifies, destroys 2-cycles, prunes high-degree vertices, and invokes Duke–Lefmann–Rödl. Therefore it would be inaccurate to market the sampling/pruning technology as new.

What was not located is a published theorem allowing *one exceptional colour* controlled only by its average degree and parameterizing the output by

$$B, \quad dB \lesssim N^{2/3}, \quad B \lesssim N^{1/15},$$

with rainbow-clique output $\Omega(N^{1/3}(\log B)^{1/3})$. The appropriate novelty claim is thus “a parameterized extraction/adaptation of the AJMP–DLR mechanism”, not “a new sparsification method”.

The broader locally bounded rainbow literature, such as Kim–Kühn–Kupavskii–Osthus [5], concerns rainbow spanning structures and approximate decompositions under local/global colour bounds. It does not provide the one-exceptional-colour clique lemma above.

8.3 Book-number interpolation

Finding: no direct precedent located. Book Ramsey theory is extensive; for example, Conlon [7] determines the asymptotic Ramsey number of large books, and subsequent work studies off-diagonal and random-book Ramsey problems. Those results concern red/blue (or few-colour) Ramsey numbers of the book graph itself. The searches located no result using

$$\beta(\chi) = \max_{xy} |N_{\chi(xy)}(x) \cap N_{\chi(xy)}(y)|$$

as a structural parameter for an arbitrary-colour canonical/Erdős–Rado problem, nor an interpolation of the form in Theorem 5.2. The application therefore appears substantially more novel than the underlying AJMP sparsification mechanism.

8.4 Fractional self-neighbourhood theorem

Finding: no direct canonical-Ramsey precedent located; adjacent fractional-colouring theory exists. There is substantial literature on fractional colouring under local graph constraints, including Pirot–Sereni and Kelly–Postle [6]. Those papers study fractional chromatic bounds and local demands, not off-diagonal canonical Ramsey numbers. Searches for “fractional chromatic” together with “canonical Ramsey”, “Erdős–Rado”, “lexical”, “rainbow clique”, and “monochromatic neighbourhood” did not locate a theorem using $\chi_f(G_a[N_a(v)])$ to drive a canonical-Ramsey induction. Theorem 6.2 therefore appears new as a reduction/application, although its proof uses the elementary inequality $\alpha(H) \geq |V(H)|/\chi_f(H)$ and should not be presented as a new result in fractional-colouring theory.

8.5 Crucial 2026 update: the old general conjecture is false

The most important literature update is [4]. For $c \geq a^{1+\eta}$ they prove

$$\text{ER}(a, b, c) \geq c^{\Omega_\eta(ab)}$$

and explicitly state that this disproves Araujo–Peng Conjecture 19. Since the exponent in the lower bound grows linearly with a for fixed $b = 4$, the same theorem also precludes the universal fixed- m formulation of Araujo–Peng Conjecture 20 for all sufficiently large m . It does not settle the exceptional smallest case $a = b = 4$.

This changes the best narrative for the present package:

- Theorem 1.3 is an unconditional advance on the still-open $\text{ER}(4, 4, r)$ problem.
- Theorem 5.2 identifies a broad book-sparse regime in which the formerly conjectured $r^6/(\log r)^2$ scale is valid.
- Theorem 6.2 identifies a hereditary local-complexity regime in which the formerly conjectured general exponent remains valid despite the unrestricted conjecture being false.

8.6 Novelty confidence table

Statement	Current confidence	Closest located precedent / caveat
Theorem A	High	Latest 2026 off-diagonal ER paper still quotes $O(r^7/\log^2 r)$ as known best.
B-Lemma, statement	Medium–high	AJMP contains essentially the same conflict-hypergraph machinery; no one-exceptional-colour parameterized theorem located.
Book interpolation	High	Book-Ramsey literature studies book graphs, not book number as a canonical-Ramsey restriction.
Fractional reduction C	Medium–high	Fractional-colouring local-demand theory is extensive; no self-neighbourhood canonical-Ramsey induction located. Proof is elementary once formulated.

Bibliographic caution. This search covered arXiv/current preprints, publisher pages, full-text web indices, and the principal references/citation chains found from Araujo–Peng, AJMP, and recent canonical-Ramsey papers. It does not certify mathematical novelty in the formal sense; before submission, the authors should additionally run MathSciNet/zbMATH/Google Scholar citation-forward searches from [2, 1, 4] and ask one or two specialists in canonical Ramsey theory whether they know an unpublished overlap.

9 Suggested manuscript positioning

The strongest present paper architecture is not “three unrelated theorems”. It is:

1. Unconditional progress on the surviving smallest problem:

$$\text{ER}(4, 4, r) = O(r^7/(\log r)^3).$$

2. A reusable exceptional-colour mechanism: Lemma 4.1, explicitly credited as an AJMP–DLR adaptation.

3. A structural sharp regime: the monochromatic-book interpolation, recovering $O(r^6/(\log r)^2)$ under $\beta \leq r^{2-\varepsilon}$.

4. A second structural salvage theorem: bounded local fractional chromatic number restores the old conjectured exponent for all m, ℓ in that restricted class.

5. Updated obstruction picture: after the 2026 counterexample, distinguish the false unrestricted large- m conjecture from the still-open $m = 4$ problem.

This package is mathematically stronger and more current than a short note containing only Theorem 1.3. The next technical target, if one wants to strengthen it before journal submission, is to improve the μ_2 control in Section 4, or to exploit the triangle-free structure available when $m = 4$ inside the DLR route.

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